Google I/O Extended

Reasoning Over Earth Engine.

An AI Agent for Lake Victoria's Flood Risk.

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Data Scientist
GDG EE Organizer



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THE PROBLEM.

Climate data is everywhere. Answers aren't.

Seasonal Impact

Lake Victoria's shores flood every season; Nyando and Budalangi flood repeatedly.

Complexity Barrier

The satellite data exists - but reading it needs experts.

Operational Delay

Decisions are slow when every question means hours of GIS work.

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OUR SOLUTION

Ask. Get a grounded answer.

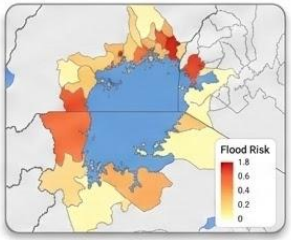
'Which sub-counties around Lake Victoria face the highest flood risk?

Gemini

Data Sources

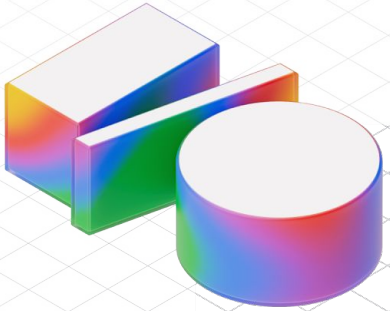
Satellite Population Reports

Results Panel



1. Muhoroni **0.80**
2. Nyando **0.79**
3. Seme **0.76**

~287,000 people exposed in Muhoroni.



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WHAT'S NEW (Google I/O 2026)

Built on Google I/O 2026

We used directly:

- **Gemini 3.5 Flash** - the new model; fast enough for live agent loops *(announcement #1)*.
- **AGENTS.md** - New markdown formats Google announced for defining an agent *(#61)*.
- **SKILL.md** - Markdown for defining Skills *(#61)*.

Inspired by Google I/O 2026

Built ourselves:

- **Agentic tool-calling + sub-agents** - expert tools echoing agent-harness primitives
- **Managed-Agents-style orchestration** - inspired by Managed Agents (#60).
- **Grounded retrieval (RAG)** - answers citing real Kenyan documents or sources.

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METHODOLOGY.

It starts with a question...



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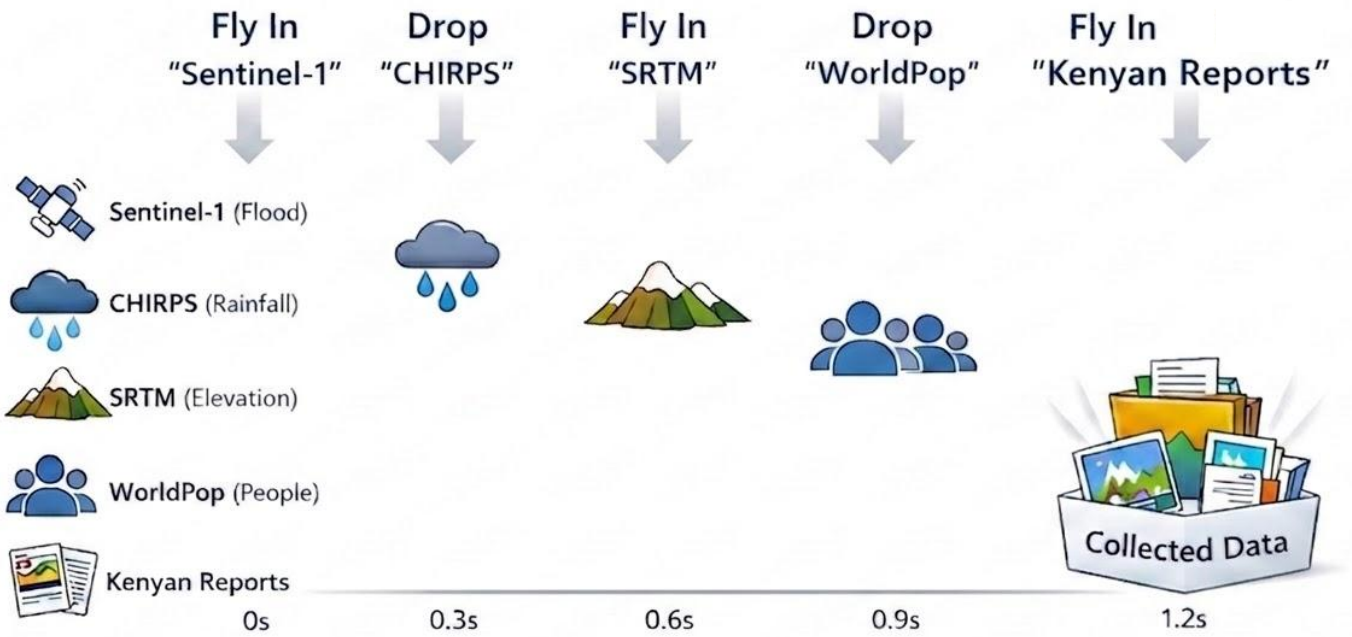
Gemini picks the right tools



The agent decides what it needs.

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Pulling real-world signals



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A transparent risk score – not a black box



Right place, right level

Two corrections that matter

LEFT PANEL

BEFORE vs AFTER

County-level results

- Kisumu
- Busia
- Migori

'Too coarse for the user's question.'

↓ Administrative correction

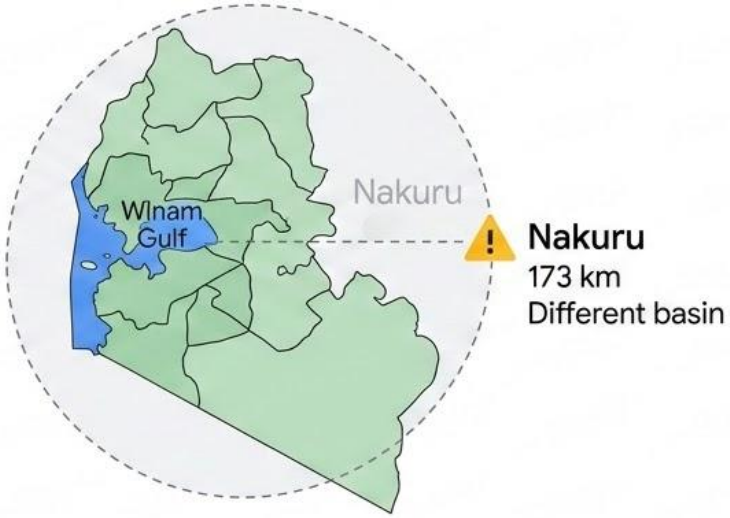
Sub-county results

- Nyando
- Nyakach
- Budalangi
- Seme

'Correct administrative level.'

RIGHT PANEL

GEOGRAPHIC GUARDRAIL



Queries are constrained to the Lake Victoria basin.
Locations outside the basin are automatically excluded.

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From data to decision

Ranked sub-counties · Interactive map · Recommendations with citations

DECISION DASHBOARD

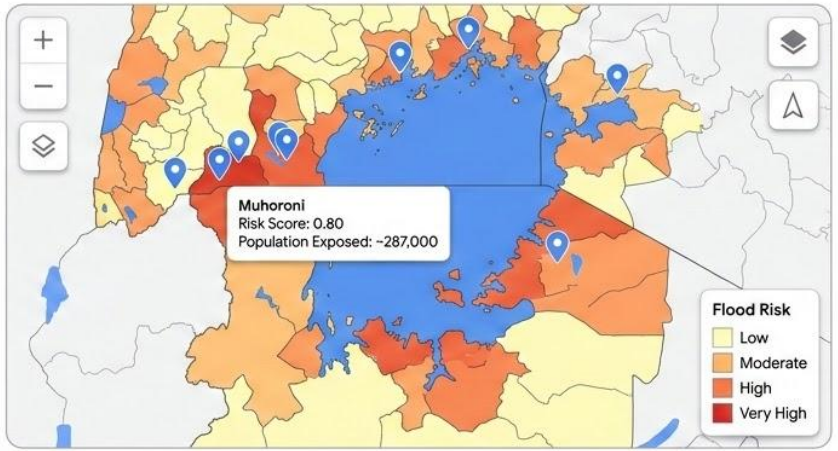
Highest Flood Risk Sub-counties

Rank	Sub-county	Risk Score	Population Exposed
1.	Muhoroni	0.80	~287,000
2.	Nyando	0.79	~251,000
3.	Seme	0.76	~194,000

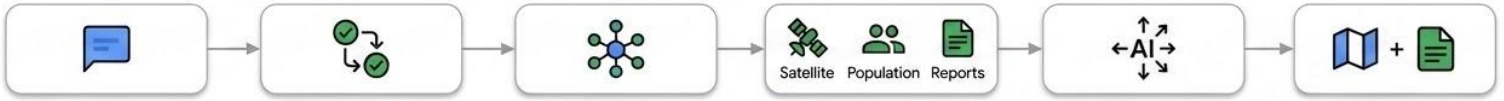
AI Recommendation

Prioritize flood preparedness and early warning deployment in Muhoroni and Nyando due to the combination of high flood susceptibility and large exposed populations.

Satellite Data · Population · Government Reports



SYSTEM ARCHITECTURE



Question
User asks a grounded question

Planner
Interprets intent

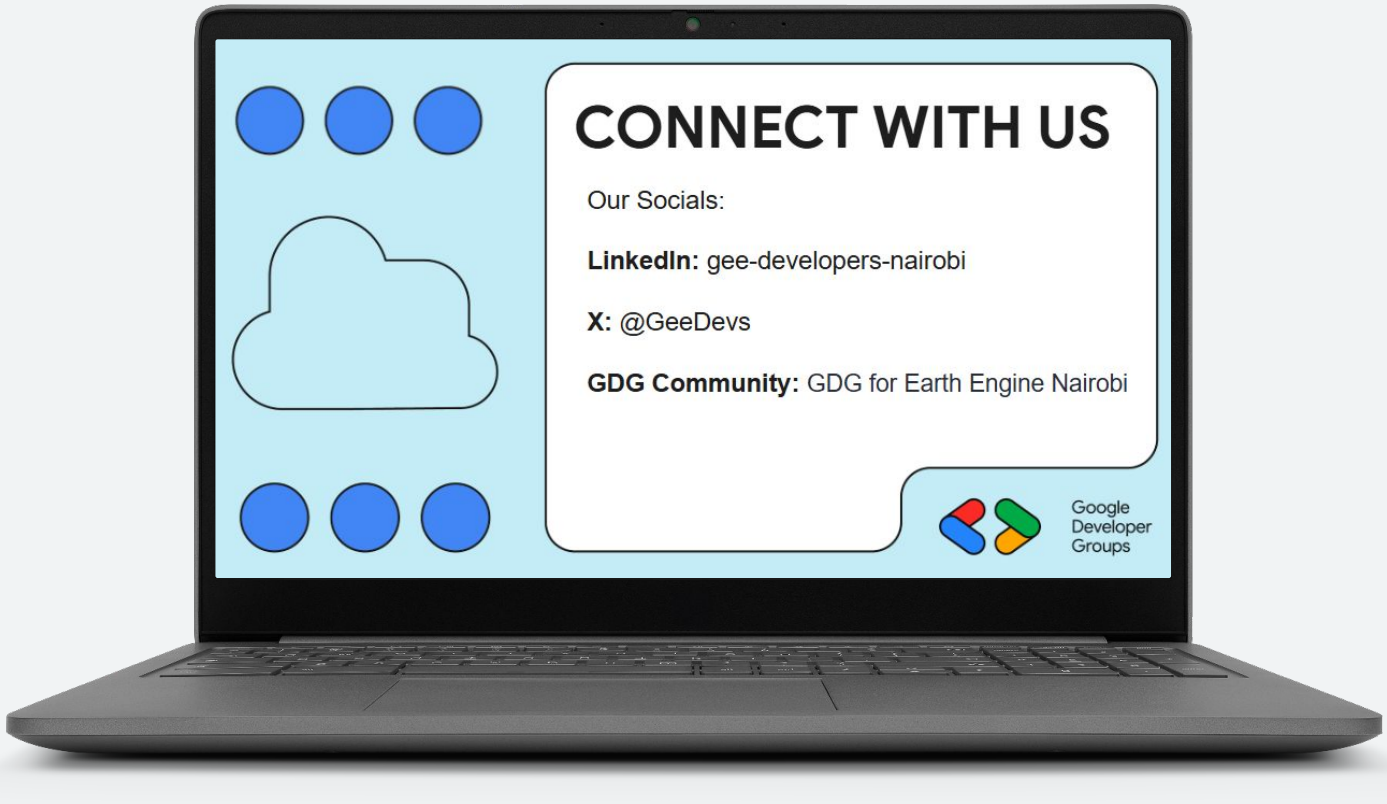
Orchestrator
Coordinates execution

Tools
Retrieves spatial, demographic and report data

Fusion
Combines evidence into one result

Map + Answer
Produces grounded recommendations with citations


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M
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